



SW Pilot 학습보고서 <과정 8>

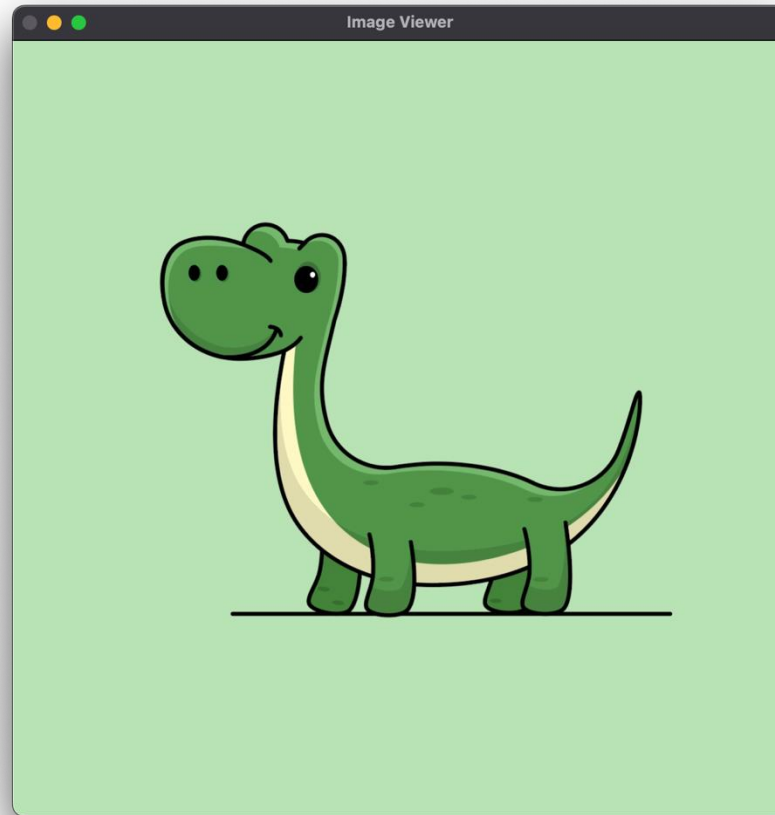
파이썬 5팀 황문규

문제 1. 여러 개의 눈 - 코드

```
1 # 과정 8 - (문제1) 여러개의 눈
2
3 import cv2
4 import os
5
6 class MediaViewer:
7     def __init__(self):
8         self.image_folder = 'src/Process8/8-1/images'
9         self.video_folder = 'src/Process8/8-1/videos'
10
11     def show_camera(self):
12         cap = cv2.VideoCapture(0)
13
14         # 해상도 설정
15         cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
16         cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
17
18         if not cap.isOpened():
19             print('카메라를 열 수 없습니다.')
20             return
21
22         while True:
23             ret, frame = cap.read()
24             if not ret:
25                 print('프레임을 읽을 수 없습니다.')
26                 break
27
28             cv2.imshow('Camera Output', frame)
29
30             if cv2.waitKey(33) & 0xFF == ord('q'):
31                 break
32
33         cap.release()
34         cv2.destroyAllWindows()
35
36     def show_images(self):
37         if not os.path.isdir(self.image_folder):
38             print(f'이미지 폴더가 존재하지 않습니다: {self.image_folder}')
39             return
40
41         for file_name in os.listdir(self.image_folder):
42             if not file_name.lower().endswith(('.png', '.jpg', '.jpeg', '.bmp', '.tif', '.tiff')):
43                 continue
44
45             file_path = os.path.join(self.image_folder, file_name)
46             image = cv2.imread(file_path)
47
48             if image is None:
49                 print(f'이미지를 열 수 없습니다: {file_path}')
50                 continue
51
52             cv2.imshow('Image Viewer', image)
53             cv2.waitKey(0)
54             cv2.destroyAllWindows()
```

```
1 # [보너스 과제] mp4 영상 출력
2 def show_videos(self):
3     if not os.path.isdir(self.video_folder):
4         print(f'비디오 폴더가 존재하지 않습니다: {self.video_folder}')
5         return
6
7     for file_name in os.listdir(self.video_folder):
8         if not file_name.lower().endswith('.mp4'):
9             continue
10
11         file_path = os.path.join(self.video_folder, file_name)
12         cap = cv2.VideoCapture(file_path)
13
14         if not cap.isOpened():
15             print(f'비디오를 열 수 없습니다: {file_path}')
16             continue
17
18         while True:
19             ret, frame = cap.read()
20             if not ret:
21                 break
22
23             cv2.imshow('Video Viewer', frame)
24
25             if cv2.waitKey(33) & 0xFF == ord('q'):
26                 break
27
28         cap.release()
29         cv2.destroyAllWindows()
30
31 if __name__ == '__main__':
32     viewer = MediaViewer()
33
34     print('1. 카메라 출력')
35     viewer.show_camera()
36
37     print('2. 이미지 출력')
38     viewer.show_images()
39
40     # [보너스 과제] mp4 영상 출력
41     print('3. [보너스] 비디오 출력')
42     viewer.show_videos()
```

문제 1. 여러 개의 눈 - 구현화면



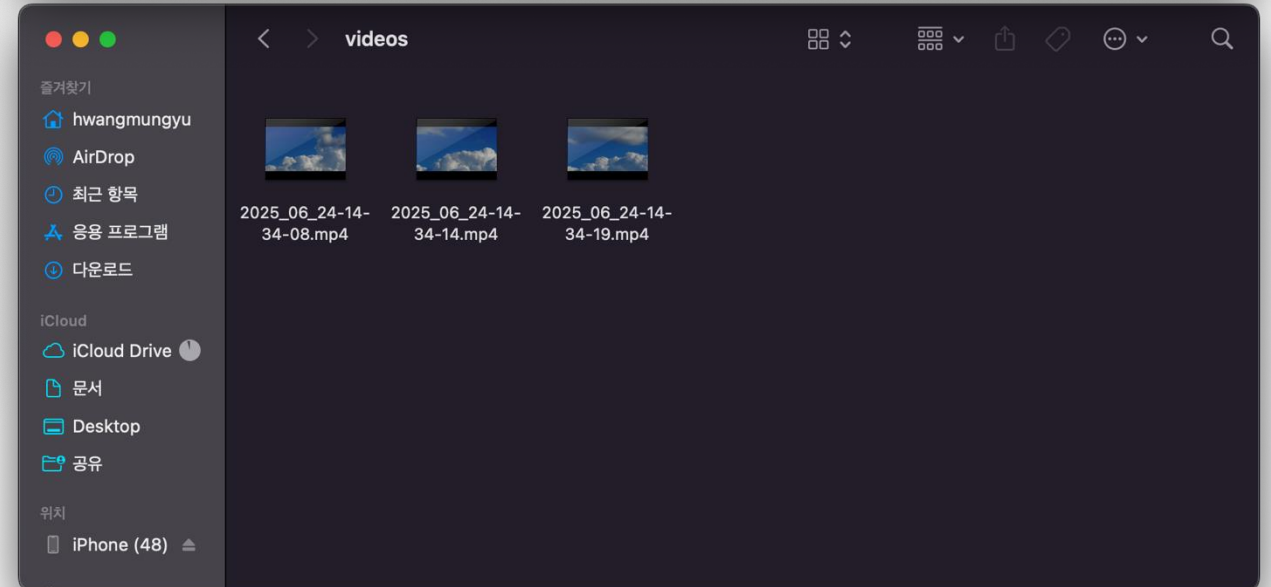
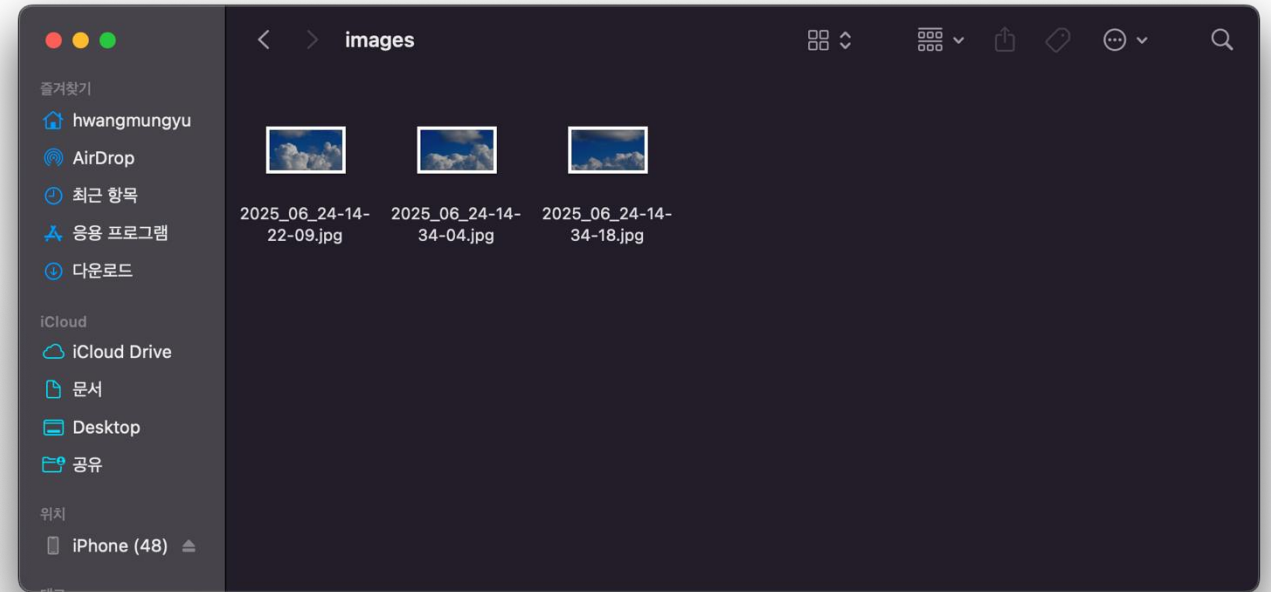
문제 2. 클립과 사진을 모아보자 - 코드

```
1 # 과제 8 - (문제2) 클립과 사진을 모아보자
2
3 import cv2
4 import os
5 from datetime import datetime
6
7
8 class VideoPlayer:
9     def __init__(self, file_path):
10         self.file_path = file_path
11         self.capture = cv2.VideoCapture(file_path)
12         self.writer = None
13         self.recording = False
14         self.codec_index = 0 # [보너스 과제] 코덱 전환용
15         self.codec_list = ['XVID', 'mp4v'] # [보너스 과제] 두 가지 코덱 사용
16
17         if not self.capture.isOpened():
18             print('비디오 파일을 열 수 없습니다.')
19             exit(1)
20
21         self.frame_width = int(self.capture.get(cv2.CAP_PROP_FRAME_WIDTH))
22         self.frame_height = int(self.capture.get(cv2.CAP_PROP_FRAME_HEIGHT))
23         self.fps = int(self.capture.get(cv2.CAP_PROP_FPS))
24
25         if self.fps == 0:
26             self.fps = 30 # fallback
27
28     def get_filename(self, ext):
29         now = datetime.now()
30         timestamp = now.strftime('%Y_%m_%d-%H-%M-%S')
31         return f'{timestamp}.{ext}'
32
33     def start_recording(self):
34         file_name = self.get_filename('mp4')
35         fourcc = cv2.VideoWriter_fourcc(*self.codec_list[self.codec_index])
36         self.writer = cv2.VideoWriter('src/Process8/8-2/videos/' + file_name, fourcc, self.fps,
37                                     (self.frame_width, self.frame_height))
38         self.recording = True
39         print(f'녹화 시작: {file_name} (코덱: {self.codec_list[self.codec_index]})')
40
41     def stop_recording(self):
42         if self.writer:
43             self.writer.release()
44             self.writer = None
45             print('녹화 중지')
46         self.recording = False
47
48     def capture_image(self, frame):
49         file_name = self.get_filename('jpg')
50         cv2.imwrite('src/Process8/8-2/images/' + file_name, frame)
51         print(f'이미지 캡처 완료: {file_name}')
```

```
1     def play(self):
2         print('영상 재생을 시작합니다.')
3         print('ESC: 종료, Ctrl+Z: 캡처, Ctrl+X: 녹화 시작, Ctrl+C: 녹화 중지')
4
5         while True:
6             ret, frame = self.capture.read()
7             if not ret:
8                 break
9
10            cv2.imshow('Video Player', frame)
11
12            key = cv2.waitKey(33) & 0xFF
13
14            # ESC 키
15            if key == 27:
16                break
17
18            # Ctrl+Z: 26 (0x1A)
19            elif key == 26:
20                self.capture_image(frame)
21
22            # Ctrl+X: 24 (0x18)
23            elif key == 24:
24                if not self.recording:
25                    self.start_recording()
26
27            # Ctrl+C: 3 (0x03)
28            elif key == 3:
29                if self.recording:
30                    self.stop_recording()
31
32            if self.recording and self.writer:
33                self.writer.write(frame)
34
35            self.capture.release()
36            if self.writer:
37                self.writer.release()
38            cv2.destroyAllWindows()
39
40
41 if __name__ == '__main__':
42     video_path = 'src/Process8/8-1/videos/1.mp4'
43
44     if not os.path.exists(video_path):
45         print(f'파일이 존재하지 않습니다: {video_path}')
46     else:
47         player = VideoPlayer(video_path)
48         player.play()
```


문제 2. 클립과 사진을 모아보자 - 구현화면

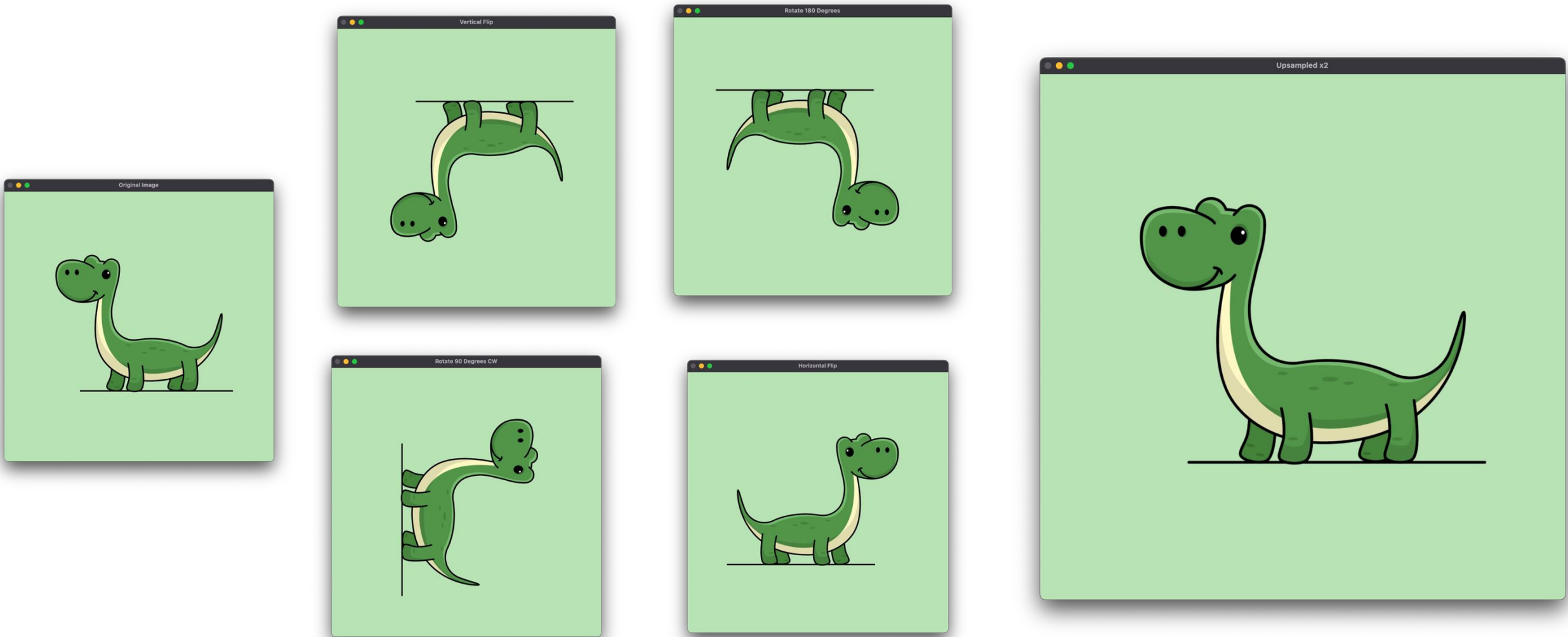
```
swpilotPyhon5 — zsh — 100x30
((swpilot) hwangmungyu@hwangmungyuui-MacBookPro swpilotPyhon5 % python src/Process8/8-2/main.py
영상 재생을 시작합니다.
ESC: 종료, Ctrl+Z: 캡처, Ctrl+X: 녹화 시작, Ctrl+C: 녹화 중지
이미지 캡처 완료: 2025_06_24-14-34-04.jpg
OpenCV: FFMPEG: tag 0x44495658/'XVID' is not supported with codec id 12 and format 'mp4 / MP4 (MPEG-4 Part 14)'
OpenCV: FFMPEG: fallback to use tag 0x7634706d/'mp4v'
녹화 시작: 2025_06_24-14-34-08.mp4 (코덱: XVID)
녹화 중지
OpenCV: FFMPEG: tag 0x44495658/'XVID' is not supported with codec id 12 and format 'mp4 / MP4 (MPEG-4 Part 14)'
OpenCV: FFMPEG: fallback to use tag 0x7634706d/'mp4v'
녹화 시작: 2025_06_24-14-34-14.mp4 (코덱: XVID)
녹화 중지
이미지 캡처 완료: 2025_06_24-14-34-18.jpg
OpenCV: FFMPEG: tag 0x44495658/'XVID' is not supported with codec id 12 and format 'mp4 / MP4 (MPEG-4 Part 14)'
OpenCV: FFMPEG: fallback to use tag 0x7634706d/'mp4v'
녹화 시작: 2025_06_24-14-34-19.mp4 (코덱: XVID)
녹화 중지
((swpilot) hwangmungyu@hwangmungyuui-MacBookPro swpilotPyhon5 %
```



문제 3. 사진들의 리터칭 - 코드

```
1 # 과정 8 - (문제3) 사진들의 리터칭
2
3 import cv2
4 import os
5
6 class ImageTransformer:
7     def __init__(self, file_path):
8         self.file_path = file_path
9         self.image = cv2.imread(file_path)
10
11     if self.image is None:
12         print(f'이미지를 불러올 수 없습니다: {file_path}')
13         exit(1)
14
15     def show_image(self, title, image):
16         cv2.imshow(title, image)
17         cv2.waitKey(0)
18         cv2.destroyAllWindows()
19
20     def show_original(self):
21         self.show_image('Original Image', self.image)
22
23     def flip_vertical(self):
24         flipped = cv2.flip(self.image, 0)
25         self.show_image('Vertical Flip', flipped)
26
27     def flip_horizontal(self):
28         flipped = cv2.flip(self.image, 1)
29         self.show_image('Horizontal Flip', flipped)
30
31     def rotate_90_clockwise(self):
32         rotated = cv2.rotate(self.image, cv2.ROTATE_90_CLOCKWISE)
33         self.show_image('Rotate 90 Degrees CW', rotated)
34
35     def rotate_180(self):
36         rotated = cv2.rotate(self.image, cv2.ROTATE_180)
37         self.show_image('Rotate 180 Degrees', rotated)
38
39     def upsample_image(self):
40         # [보너스 과제] 이미지 크기를 2배로 업샘플링
41         upsampled = cv2.resize(self.image, None, fx=2.0, fy=2.0, interpolation=cv2.INTER_LINEAR)
42         self.show_image('Upsampled x2', upsampled)
43
44
45 if __name__ == '__main__':
46     image_path = 'src/Process8/8-1/images/2.jpg'
47
48     if not os.path.exists(image_path):
49         print(f'파일이 존재하지 않습니다: {image_path}')
50     else:
51         transformer = ImageTransformer(image_path)
52         transformer.show_original()
53         transformer.flip_vertical()
54         transformer.flip_horizontal()
55         transformer.rotate_90_clockwise()
56         transformer.rotate_180()
57         transformer.upsample_image()
```

문제 3. 사진들의 리터칭 - 구현화면

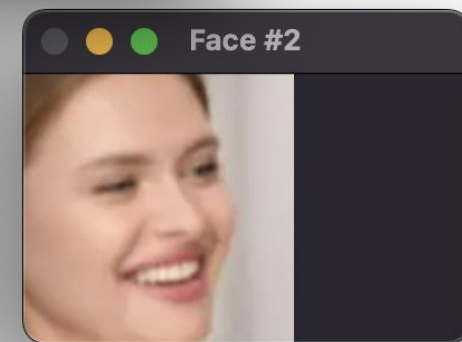
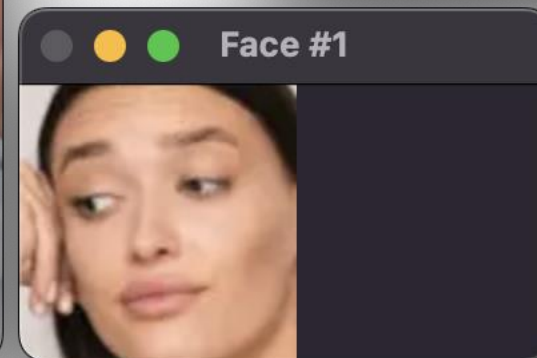
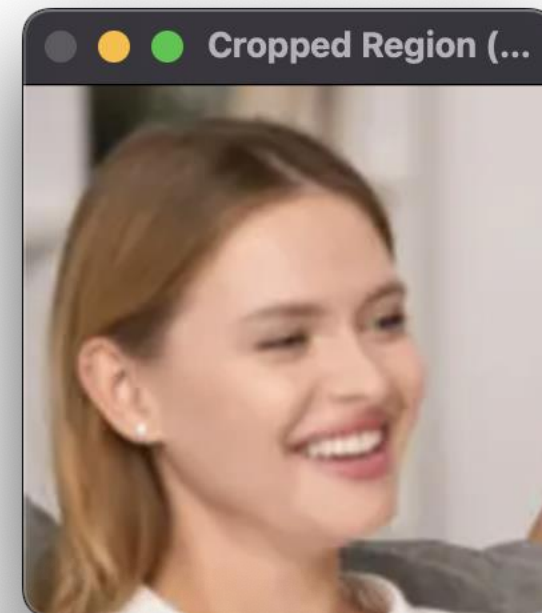


문제 4. 이미지의 크기를 바꾸면 다시 보이는 것들 - 코드

```
1 # 과정 8 - (문제4) 이미지의 크기를 바꾸면 다시 보이는 것들
2 import cv2
3 import os
4 import copy
5
6 class ImageProcessor:
7     def __init__(self, file_path):
8         self.file_path = file_path
9         self.image = cv2.imread(file_path)
10
11     if self.image is None:
12         print(f'이미지를 불러올 수 없습니다: {file_path}')
13         exit(1)
14
15     def show_image(self, title, image):
16         cv2.imshow(title, image)
17         cv2.waitKey(0)
18         cv2.destroyAllWindows()
19
20     def resize_absolute(self, width, height):
21         resized = cv2.resize(self.image, (width, height))
22         self.show_image(f'Resized to {width}x{height}', resized)
23
24     def resize_relative(self, fx, fy):
25         resized = cv2.resize(self.image, None, fx=fx, fy=fy, interpolation=cv2.INTER_LINEAR)
26         self.show_image(f'Scaled by fx={fx}, fy={fy}', resized)
27
28     def crop_region(self, x, y, w, h):
29         region = self.image[y:y+h, x:x+w]
30         copied = copy.deepcopy(region) # deep copy
31         self.show_image('Cropped Region (Deep Copy)', copied)
```

```
1 # [보너스 과제] 얼굴 개별 추출
2 def detect_and_crop_faces(self):
3     gray = cv2.cvtColor(self.image, cv2.COLOR_BGR2GRAY)
4
5     face_cascade = cv2.CascadeClassifier(cv2.data.harcascades + 'haarcascade_frontalface_default.xml')
6
7     faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5)
8
9     print(f'검지된 얼굴 수: {len(faces)}')
10
11     for idx, (x, y, w, h) in enumerate(faces):
12         face = self.image[y:y+h, x:x+w]
13         copied_face = copy.deepcopy(face)
14         self.show_image(f'Face #{idx + 1}', copied_face)
15
16 if __name__ == '__main__':
17     image_path = 'src/Process8/8-1/images/5.png'
18
19     if not os.path.exists(image_path):
20         print(f'파일이 존재하지 않습니다: {image_path}')
21     else:
22         processor = ImageProcessor(image_path)
23
24         # 절대 크기로 리사이즈
25         processor.resize_absolute(640, 480)
26         processor.resize_absolute(1024, 768)
27
28         # 상대 크기로 리사이즈
29         processor.resize_relative(fx=0.3, fy=0.7)
30
31         # 특정 영역 잘라내기 (예: 이미지 좌상단 200x200 크기)
32         processor.crop_region(50, 50, 200, 200)
33
34 # [보너스 과제] 얼굴 개별 추출
35 processor.detect_and_crop_faces()
```


문제 4. 이미지의 크기를 바꾸면 다시 보이는 것들 - 구현화면

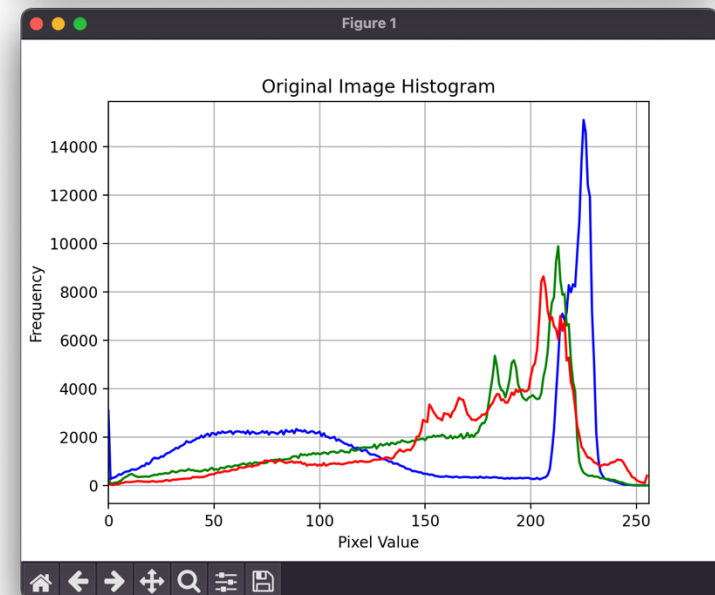
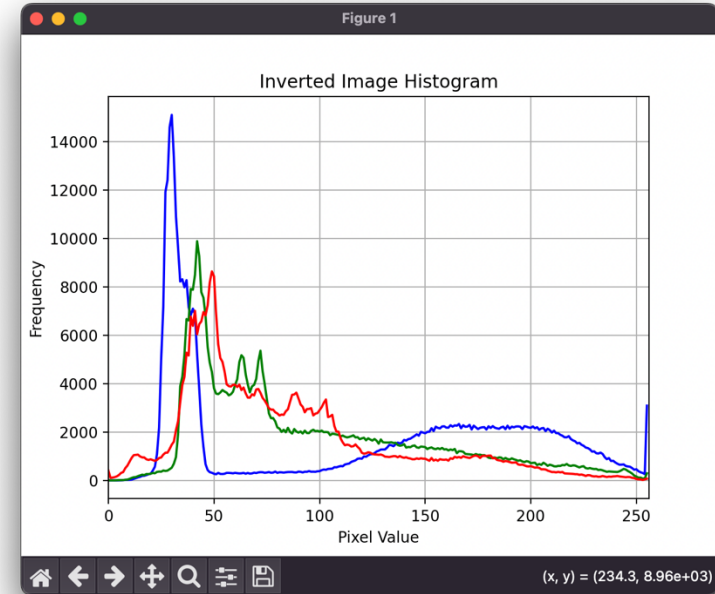


문제 5. 공간을 바꾸고 뒤집으면 보이는 것들 - 코드

```
1 # 과제 8 - (문제5) 공간을 바꾸고 뒤집으면 보이는 것들
2 import cv2
3 import os
4 import numpy as np
5 import matplotlib.pyplot as plt
6
7
8 class ImageInverter:
9     def __init__(self, file_path):
10         self.file_path = file_path
11         self.image = cv2.imread(file_path)
12
13         if self.image is None:
14             print(f'이미지를 불러올 수 없습니다: {file_path}')
15             exit(1)
16
17     def show_image(self, title, image):
18         cv2.imshow(title, image)
19         cv2.waitKey(0)
20         cv2.destroyAllWindows()
21
22     def convert_to_gray(self):
23         gray = cv2.cvtColor(self.image, cv2.COLOR_BGR2GRAY)
24         self.show_image('Grayscale Image', gray)
25         return gray
26
27     def invert_image(self):
28         inverted = cv2.bitwise_not(self.image)
29         self.show_image('Inverted Image', inverted)
30         return inverted
31
```

```
1 # [보너스 과제] - 히스토그램 출력
2 def plot_histogram(self, image, title):
3     if len(image.shape) == 3:
4         # 컬러 이미지
5         colors = ('b', 'g', 'r')
6         for i, color in enumerate(colors):
7             hist = cv2.calcHist([image], [i], None, [256], [0, 256])
8             plt.plot(hist, color=color)
9     else:
10        # 흑백 이미지
11        hist = cv2.calcHist([image], [0], None, [256], [0, 256])
12        plt.plot(hist, color='gray')
13
14    plt.title(title)
15    plt.xlabel('Pixel Value')
16    plt.ylabel('Frequency')
17    plt.xlim([0, 256])
18    plt.grid(True)
19    plt.show()
20
21
22 if __name__ == '__main__':
23     image_path = 'src/Process8/8-1/images/4.jpg'
24
25     if not os.path.exists(image_path):
26         print(f'파일이 존재하지 않습니다: {image_path}')
27     else:
28         inverter = ImageInverter(image_path)
29
30         # 색상 전환: RGB → GRAY
31         gray_image = inverter.convert_to_gray()
32
33         # 역상 이미지
34         inverted_image = inverter.invert_image()
35
36         # [보너스 과제] - 히스토그램 출력
37         inverter.plot_histogram(inverter.image, 'Original Image Histogram') # 원본
38         inverter.plot_histogram(inverted_image, 'Inverted Image Histogram') # 역상
```

문제 5. 공간을 바꾸고 뒤집으면 보이는 것들 - 구현화면



문제 6. 마음이 아려오는 이미지들... - 코드

```
1 # 과정 8 - (문제6) 마음이 아려오는 이미지들...
2 import cv2
3 import os
4
5 class EdgeAndBlurProcessor:
6     def __init__(self, binary_image_path, blur_image_path):
7         self.binary_image_path = binary_image_path
8         self.blur_image_path = blur_image_path
9
10        self.image = cv2.imread(binary_image_path)
11        self.blur_image = cv2.imread(blur_image_path)
12
13        if self.image is None:
14            print(f'이진화용 이미지를 불러올 수 없습니다: {binary_image_path}')
15            exit(1)
16
17        if self.blur_image is None:
18            print(f'흐림 효과용 이미지를 불러올 수 없습니다: {blur_image_path}')
19            exit(1)
20
21        def show_image(self, title, image):
22            cv2.imshow(title, image)
23            cv2.waitKey(0)
24            cv2.destroyAllWindows()
25
26        def convert_to_gray(self):
27            gray = cv2.cvtColor(self.image, cv2.COLOR_BGR2GRAY)
28            self.show_image('Grayscale', gray)
29            return gray
30
31        def binary_threshold(self, gray_image):
32            _, binary = cv2.threshold(gray_image, 127, 255, cv2.THRESH_BINARY)
33            self.show_image('Binary Image', binary)
34
35        def detect_edges(self):
36            gray = cv2.cvtColor(self.image, cv2.COLOR_BGR2GRAY)
37
38            # Sobel (x + y 방향)
39            sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
40            sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
41            sobel = cv2.magnitude(sobelx, sobely)
42            sobel = cv2.convertScaleAbs(sobel)
43            self.show_image('Sobel Edge', sobel)
44
45            # Laplacian
46            laplacian = cv2.Laplacian(gray, cv2.CV_64F)
47            laplacian = cv2.convertScaleAbs(laplacian)
48            self.show_image('Laplacian Edge', laplacian)
```

```
1 # Canny
2 canny = cv2.Canny(gray, 100, 200)
3 self.show_image('Canny Edge', canny)
4
5 def apply_blur(self):
6     blurred = cv2.GaussianBlur(self.blur_image, (15, 15), 0)
7     self.show_image('Blurred Image', blurred)
8
9 def blur_region_only(self):
10     # [보너스 과제] 이미지의 일부분만 흐림 적용 (중앙 사각형)
11     result = self.blur_image.copy()
12
13     h, w = result.shape[:2]
14     x1 = w // 3
15     y1 = h // 3
16     x2 = x1 + w // 3
17     y2 = y1 + h // 3
18
19     roi = result[y1:y2, x1:x2]
20     blurred_roi = cv2.GaussianBlur(roi, (31, 31), 0)
21     result[y1:y2, x1:x2] = blurred_roi
22
23     self.show_image('Partial Blur (보너스)', result)
24
25 if __name__ == '__main__':
26     bin_img_path = 'src/Process8/8-1/images/4.jpg'
27     blur_img_path = 'src/Process8/8-1/images/4.jpg'
28
29     if not os.path.exists(bin_img_path) or not os.path.exists(blur_img_path):
30         print('필요한 이미지 파일이 존재하지 않습니다.')
31     else:
32         processor = EdgeAndBlurProcessor(bin_img_path, blur_img_path)
33
34         # 그레이스케일 → 이진화
35         gray_img = processor.convert_to_gray()
36         processor.binary_threshold(gray_img)
37
38         # 가장자리 검출
39         processor.detect_edges()
40
41         # 흐림 효과 전체
42         processor.apply_blur()
43
44         # [보너스] 특정 영역만 흐림
45         processor.blur_region_only()
```

문제 6. 마음이 내려오는 이미지들... - 구현화면

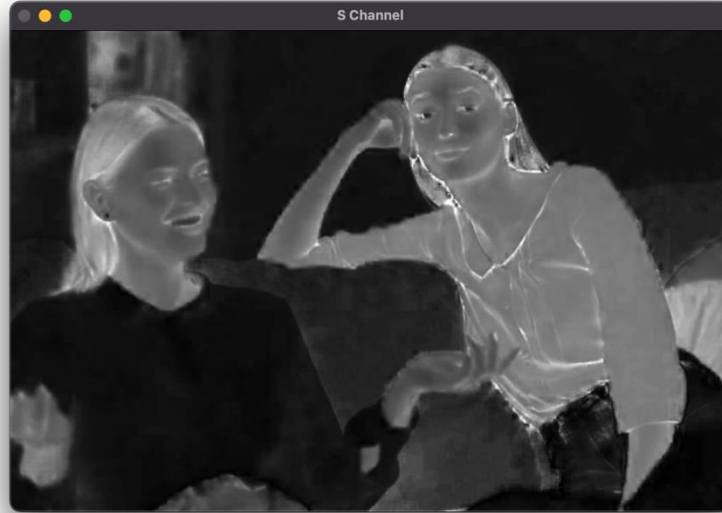


문제 7. 색에도 성분이 있다. - 코드

```
1 # 과정 8 - (문제7) 색에도 성분이 있다.
2 import cv2
3 import os
4
5 class HsvAnalyzer:
6     def __init__(self, file_path):
7         self.file_path = file_path
8         self.image = cv2.imread(file_path)
9
10        if self.image is None:
11            print(f'이미지를 불러올 수 없습니다: {file_path}')
12            exit(1)
13
14        def show_image(self, title, image):
15            cv2.imshow(title, image)
16            cv2.waitKey(0)
17            cv2.destroyAllWindows()
18
19        def convert_to_hsv(self):
20            hsv_image = cv2.cvtColor(self.image, cv2.COLOR_BGR2HSV)
21            return hsv_image
22
23        def split_and_show_hsv(self, hsv_image):
24            h_channel, s_channel, v_channel = cv2.split(hsv_image)
25
26            self.show_image('H Channel', h_channel)
27            self.show_image('S Channel', s_channel)
28            self.show_image('V Channel', v_channel)
```

```
1 # [보너스 과제] - 채널 분리 기법
2 def compare_with_bgr_split(self):
3     b_channel, g_channel, r_channel = cv2.split(self.image)
4
5     self.show_image('B Channel (BGR)', b_channel)
6     self.show_image('G Channel (BGR)', g_channel)
7     self.show_image('R Channel (BGR)', r_channel)
8
9 if __name__ == '__main__':
10     image_path = 'src/Process8/8-1/images/5.png'
11
12     if not os.path.exists(image_path):
13         print(f'파일이 존재하지 않습니다: {image_path}')
14     else:
15         analyzer = HsvAnalyzer(image_path)
16
17         # HSV 변환 및 분리 출력
18         hsv_img = analyzer.convert_to_hsv()
19         analyzer.split_and_show_hsv(hsv_img)
20
21         # [보너스 과제] BGR 채널 분리 비교
22         analyzer.compare_with_bgr_split()
```

문제 7. 색에도 성분이 있다. - 구현화면



문제 8. 도형 그리기 - 코드

```
1 # 과정 8 - (문제8) 도형 그리기
2 import cv2
3 import os
4 import numpy as np
5
6 class ObjectAnnotator:
7     def __init__(self, file_path):
8         self.file_path = file_path
9         self.image = cv2.imread(file_path)
10
11     if self.image is None:
12         print(f'이미지를 불러올 수 없습니다: {file_path}')
13         exit(1)
14
15     # 사전 정의된 객체 위치와 유형 (예제용)
16     self.objects = [
17         {'name': 'Box', 'type': 'box', 'pos': (100, 100, 100, 100)}, # (x, y, w, h)
18         {'name': 'Ball', 'type': 'ball', 'pos': (250, 120, 80, 80)},
19         {'name': 'Cone', 'type': 'cone', 'pos': (400, 130, 100, 100)},
20     ]
21
22     def annotate_objects(self):
23         for obj in self.objects:
24             name = obj['name']
25             shape_type = obj['type']
26             x, y, w, h = obj['pos']
27
28             center_x = x + w // 2
29             center_y = y + h // 2
30
31             # [보너스 과제] - 도형 그리기 (빨간색, 물품 종류에 따라 다른 모양으로 구분)
32             color = (0, 0, 255)
33             thickness = 2
```

```
1         if shape_type == 'box':
2             cv2.rectangle(self.image, (x, y), (x + w, y + h), color, thickness)
3
4         elif shape_type == 'ball':
5             radius = min(w, h) // 2
6             cv2.circle(self.image, (center_x, center_y), radius, color, thickness)
7
8         elif shape_type == 'cone':
9             point1 = (x + w // 2, y)
10            point2 = (x, y + h)
11            point3 = (x + w, y + h)
12            triangle = np.array([point1, point2, point3])
13            cv2.polylines(self.image, [triangle], isClosed=True, color=color, thickness=thickness)
14
15            # 텍스트 위치 및 연결선
16            text_x = x + w + 40
17            text_y = y + 20
18            cv2.putText(self.image, name, (text_x, text_y), cv2.FONT_HERSHEY_SIMPLEX, 0.6, color, 2)
19
20            line_start = (center_x, center_y)
21            line_end = (text_x, text_y)
22            cv2.line(self.image, line_start, line_end, color, 1)
23
24     def show_result(self):
25         cv2.imshow('Annotated Image', self.image)
26         cv2.waitKey(0)
27         cv2.destroyAllWindows()
28
29 if __name__ == '__main__':
30     image_path = 'src/Process8/8-1/images/6.jpg'
31
32     if not os.path.exists(image_path):
33         print(f'파일이 존재하지 않습니다: {image_path}')
34     else:
35         annotator = ObjectAnnotator(image_path)
36         annotator.annotate_objects()
37         annotator.show_result()
```

문제 8. 도형 그리기 - 구현화면

